# TrueMark

# CAPSTONE PROJECT PHASE-I

## Phase – I Report

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# Bonafide Certificate

Certified that this project report titled **"TrueMark"** is the bonafide work of "22BCE10354 Jit Surani, 22BCE10364 Rushabh Madan Wagh, 22BCE10385 Kavya, 22BCE10853 Tejas Pathak, 22BCE10914 Simarpreet Singh**"** who carried out the project work under my supervision.

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**Supervisor**

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# List of Abbreviations

| Sl. No. | Abbreviation | Full |
|---|---|---|
| 1 | LSB | Least Significant Bit |
| 2 | SHA-256 | Secure Hash Algorithm 256-bit |
| 3 | AES | Advanced Encryption Standard |
| 4 | CBC | Cipher Block Chaining |
| 5 | AEAD | Authenticated Encryption with Associated Data |
| 6 | AES-GCM | Advanced Encryption Standard - |
| 7 | CRC32 | Cyclic Redundancy Check 32 Bit |
| 8 | C2PA | Coalition for Content Provenance and Authenticity |
| 9 | RSA | Rivest–Shamir–Adleman |
| 10 | PSNR | Peak Signal-to-Noise Ratio |

# Chapter 1: Introduction

## 1.1 Motivation

In today's digital landscape, images and documents are shared across social media platforms, messaging applications, and cloud services at an unprecedented rate. Users routinely exchange sensitive files such as identity cards, academic certificates, medical reports, legal documents, creative artworks, and proprietary designs. However, this widespread digital sharing occurs without adequate security measures, exposing individuals and organizations to significant risks including tampering, identity theft, unauthorized redistribution, privacy breaches, and exploitation by artificial intelligence systems.

Traditional image protection methods have proven insufficient in addressing contemporary security challenges. Visible watermarks are easily removed through cropping, blurring, or sophisticated editing techniques. Metadata-based protection mechanisms fail to provide robust security, as metadata can be stripped using readily available tools. Copyright enforcement procedures remain slow, manual, and reactive, often unable to prevent substantial damage before intervention occurs. Furthermore, modern artificial intelligence platforms increasingly scrape publicly available images for training datasets without explicit creator consent, resulting in loss of ownership, attribution, and creative control.

The emergence of generative AI models and large-scale image datasets has exacerbated these concerns. Content creators find their original works being used to train commercial AI systems without compensation or acknowledgment. Individuals discover their personal photographs incorporated into facial recognition systems without authorization. Organizations face intellectual property theft as proprietary visual assets are harvested and replicated by competitors. These challenges collectively demonstrate an urgent need for robust, tamper-resistant, and automated protection mechanisms for digital images—systems that can secure ownership, protect privacy, verify authenticity, and prevent unauthorized usage across platforms and applications.

TrueMark addresses these critical security gaps by implementing a comprehensive image protection framework that embeds cryptographic trust directly into digital images. Rather than relying on external metadata or easily removable visual markers, TrueMark integrates security at the pixel level through invisible watermarking, cryptographic signatures, and encryption. This approach ensures that protection travels with the image regardless of distribution channel, platform, or subsequent transformations.

The development of TrueMark is motivated by three fundamental principles:

**Protection of Ownership and Intellectual Property:** Content creators, photographers, artists, and designers require reliable technical mechanisms to prove authorship and establish chain of custody for their digital works. TrueMark provides cryptographically verifiable proof of ownership that survives image transformations and cannot be easily forged or removed.

**Security and Privacy of Sensitive Visual Content:** Personal identification documents, medical imaging, confidential business materials, and private photographs require protection both during transmission and storage. TrueMark implements military-grade encryption and

integrity verification to ensure that sensitive visual information remains confidential and unaltered throughout its lifecycle.

**Ethical and Responsible AI Development:** As artificial intelligence systems become increasingly dependent on large-scale image datasets, there is growing recognition of the need for technical measures that enable content creators to opt-out of AI training pipelines.

TrueMark aims to provide machine-readable protection signals that AI platforms can detect and respect, supporting the development of ethical AI systems that honor creator rights.

The proliferation of deepfakes, image manipulation technologies, and AI-generated synthetic media has created an environment where visual authenticity can no longer be assumed. In contexts ranging from journalism and legal proceedings to social media and personal communications, the ability to verify image provenance and detect tampering has become essential. TrueMark contributes to addressing this "trust crisis" in digital media by providing technical infrastructure for image authentication and verification.

From a practical standpoint, existing security solutions often require specialized knowledge, expensive software licenses, or complex workflows that limit accessibility for ordinary users. TrueMark is designed as a mobile-first application that makes advanced cryptographic image protection accessible to non-technical users through an intuitive interface. By lowering the barrier to entry for secure image management, TrueMark democratizes access to protection technologies previously available only to enterprises and technical specialists.

Thus, TrueMark represents a timely and necessary response to evolving threats in the digital image ecosystem. It is envisioned as a practical, user-friendly, and security-driven solution to a rapidly growing global challenge that affects individuals, creative professionals, organizations, and society at large.

## 1.2 Objective

The primary objective of TrueMark is to design and implement a secure image protection and verification system that integrates cryptographic techniques, steganographic watermarking, and symmetric encryption within a cross-platform mobile application framework. This research project aims to demonstrate the feasibility of embedding tamper-evident security mechanisms directly into digital images while maintaining usability for non-technical users.

### Major Objectives

### 1. Develop a Secure Image Protection Pipeline

The core technical objective is to implement a multi-layered protection system that combines complementary security mechanisms:

- Embed invisible, tamper-resistant watermarks into image data using steganographic techniques, specifically Least Significant Bit (LSB) modification in the spatial domain, with a roadmap for advanced frequency-domain methods (Discrete Cosine Transform and Discrete Wavelet Transform)

- Generate unique cryptographic identifiers for each protected image using secure hash functions (SHA-256) to establish content fingerprints that can detect unauthorized modifications
- Implement symmetric encryption using Advanced Encryption Standard (AES) in Cipher Block Chaining (CBC) mode with 256-bit keys to ensure confidentiality during storage and transmission, with future migration to Authenticated Encryption with Associated Data (AEAD) modes such as AES-GCM
- Design and implement payload structures that include magic headers, length fields, encrypted data, and Cyclic Redundancy Check (CRC32) integrity verification to enable robust extraction and validation

**2. Build a Cross-Platform Mobile Application**

Develop a production-quality mobile application using the Flutter framework that enables users to:

- Capture images directly through device cameras or select existing images from gallery storage
- Generate unique image identifiers through a hybrid system combining user-specific base numbers with sequential counters, ensuring global uniqueness while maintaining user traceability
- Process images through the protection pipeline with real-time feedback on operation status and completion
- Manage protected images with metadata including processing timestamps, file sizes, cryptographic hashes, and associated identifiers
- Access verification functionality to authenticate protected images and detect tampering attempts

**3. Implement Backend Infrastructure for User Management and Metadata Storage**

Deploy a scalable cloud-based backend system utilizing Firebase services to support:

- Secure user authentication through Firebase Authentication supporting email/password credentials with extensibility for additional authentication methods
- Persistent storage of user profiles, preferences, and account metadata in Cloud Firestore NoSQL database
- Atomic transaction-based unique identifier generation ensuring consistency across concurrent operations
- Storage of image metadata including protection parameters, cryptographic hashes, processing timestamps, and verification logs organized in hierarchical document collections
- Administrative interfaces for monitoring system usage, authentication attempts, and user activity through role-based access control

**4. Create a Comprehensive Verification System**

Design and implement verification mechanisms that enable users to:

- Extract embedded watermarks from protected images using password-based decryption
- Validate data integrity through CRC32 checksum verification to detect payload corruption
- Verify encryption passwords to confirm authorization for accessing protected content

- Cross-reference extracted identifiers against Firebase database records to confirm authenticity and ownership
- Receive clear, actionable feedback on verification results including success confirmations, failure reasons, and tampering detection alerts

**5. Establish Foundation for AI Protection Capabilities**

Lay the groundwork for future detection systems that would enable:

- Machine-readable protection signals embedded within image data that AI platforms could programmatically detect
- Standardized metadata formats compatible with emerging content authenticity initiatives such as Coalition for Content Provenance and Authenticity (C2PA)
- API interfaces for automated checking of image protection status prior to inclusion in training datasets
- Documentation and technical specifications to support integration with third-party platforms and services

**6. Ensure Usability and Accessibility**

Prioritize user experience through:

- Intuitive interface design following Material Design 3 principles to minimize learning curve
- Progressive disclosure of technical details, presenting simplified workflows to novice users while exposing advanced options to power users
- Real-time visual feedback during processing operations including progress indicators and status messages
- Comprehensive error handling with user-friendly error messages and recovery suggestions
- Cross-platform compatibility across Android, iOS, Web, Windows, and macOS to maximize accessibility

**7. Validate Security and Performance**

Conduct systematic evaluation of the implemented system to:

- Measure watermark robustness against common image transformations including compression, resizing, format conversion, and minor editing operations
- Assess encryption strength and key management practices against established cryptographic standards
- Evaluate system performance metrics including processing time, memory consumption, and storage requirements across different device capabilities
- Test verification accuracy including true positive rates for authentic images and false positive rates for unprotected or tampered images
- Document security assumptions, threat model, and known limitations to guide future development

**Expected Outcomes**

Upon successful completion of this capstone project, TrueMark will demonstrate:

- A functional prototype implementation of cryptographic image protection suitable for real-world deployment
- Integration of multiple security techniques (steganography, encryption, integrity checking) into a cohesive system architecture
- Practical application of cryptographic primitives within a mobile computing context
- Scalable cloud backend infrastructure capable of supporting multiple concurrent users
- User-centered design that makes advanced security technologies accessible to non-technical audiences
- A foundation for future enhancements including digital signatures, advanced watermarking, blockchain integration, and AI detection capabilities

This project contributes to the broader field of digital content protection by demonstrating practical implementation approaches, identifying design tradeoffs between security and usability, and validating the feasibility of embedding cryptographic trust mechanisms within digital images through mobile application interfaces.

# Chapter 2: Existing Work / Literature Review

Recent work consistently shows that a layered approach combining strong encryption for confidentiality with verifiable signatures for authenticity most effectively prevents unauthorized reuse of images by AI systems and adversaries in the wild. Below are the findings that directly support TrueMark's design goal: encrypt user photos end-to-end so AI tools cannot ingest or learn from them, and bind each file to a cryptographic signature so any modification is immediately detected during verification, preserving user control over sharing and provenance in practical mobile workflows.

## 2.1 A Secure Method for Image Signaturing using SHA256, RSA and Advanced Encryption Standard

The paper proposes a three-stage digital signature scheme for images that combines SHA-256 hashing, RSA public-key cryptography, and AES symmetric encryption to create and distribute a compact signature stored as a binary file alongside the image. The process hashes both the image bytes and the filename, encrypts the concatenated 512-bit hash with RSA (2048-bit modulus), and then wraps the RSA output with AES (128-bit key), producing a 2048-bit ciphertext that can later be verified by decrypting (AES→RSA) and re-hashing the received image and filename to check equality for integrity and authenticity. Experiments with tampering (renaming, grayscale conversion, Gaussian blur, and single-pixel change) show the method flags manipulations reliably; even a one-bit change alters the SHA-256 digest, and reported PSNR values for blurred and one-pixel edits demonstrate detectable deviations, while grayscale changes are also identified despite PSNR not being applicable due to dimension changes in the input vector.

## 2.2 A Study on Digital Signatures with Privacy Factor

For an image-security application, the paper's guidance supports a clean separation of concerns: use a strong hash (e.g., SHA-256) over immutable image bytes plus a defined metadata subset, sign the digest with an application or user private key (RSA/ECDSA), and verify with the corresponding public key to ensure integrity and provenance, while handling confidentiality via symmetric encryption of the image for storage and transport when needed. It also points to operational best practices: manage keys via a lightweight PKI or embedded certificates, consider blockchain or append-only logs to timestamp signatures, and choose efficient algorithms (e.g., ECDSA over prime curves) to reduce latency on mobile clients without weakening security, aligning with the survey's performance and deployment recommendations.

## 2.3 A Comprehensive Survey on Methods for Image Integrity

This survey synthesizes methods for image integrity assurance across active and passive forensics, organizing the field by manipulation type (e.g., copy-move, splicing, resampling, double JPEG) and by evidence sources such as JPEG quantization traces, CFA/PRNU sensor fingerprints, and transform-domain statistical cues, as well as modern deep-learning detectors for forgeries and deepfakes. It reviews watermarking and zero-watermarking for active integrity, and details passive pipelines including keypoint features (SIFT, LBP), artifact analysis for compression history, and CNN-based localization, mapping strengths, failure modes, and benchmarks through metrics and datasets used in recent works.

## 2.4 A Cryptographic Framework for Secure Medical Imaging in Smart Healthcare Environments

The article proposes a hybrid cryptographic framework that combines AES-128 for fast medical-image encryption with RSA-1024 for secure key exchange, aiming to protect MRI and similar imaging data in smart healthcare systems. Experiments on a large MRI dataset show that encryption is efficient (0.5–2.9 seconds per image, with faster decryption) and achieves high throughput, especially in AES CTR and OCB modes. Security analyses including entropy, NPCR, PSNR, histograms, and correlation tests indicate strong resistance to statistical and differential attacks, with encrypted images showing near-ideal randomness and greater than 99.5% pixel-change rates. Overall, the study concludes that the scheme offers a practical balance of speed and confidentiality for telemedicine and digital-health workflows, though real-world deployment would still require consideration of stronger key sizes and broader system-level security.

## 2.5 Securing Medical Images for Mobile Health Systems Using a Combined Approach of Encryption and Steganography

The chapter presents a novel encryption scheme tailored for medical image protection in mobile-health (m-health) systems, combining a symmetric cipher for bulk image data with an asymmetric key system to secure key distribution. The authors design and implement their approach specifically for mobile devices and wireless transport of medical imaging, evaluating its performance in terms of encryption and decryption speed, image fidelity, and resilience to statistical and differential attacks. Their results show that the scheme provides sufficient protection of image confidentiality while achieving acceptable processing overhead for m-health contexts, thereby enabling secure transmission and storage of medical images on mobile platforms.

## 2.6 Privacy-Preserving Biometrics Image Encryption and Digital Signature Technique Using Arnold and ElGamal

The authors present a comprehensive cryptographic model for securing biometric images (face, palmprint and iris) that integrates encryption, signature and authentication: first the image is digitally signed by a dual-key scheme (using the ElGamal Digital Signature algorithm), then it undergoes scrambling via an enhanced three-dimensional version of the Arnold Transform, followed by further encryption of transform parameters using the ElGamal Encryption algorithm. The scheme was tested on multiple biometric datasets (including the FEI face set and IIT Delhi palmprint and iris sets) and compared with existing methods; results show good randomness, resistance to various attacks and suitability for storing and transmitting sensitive biometric image data while preserving integrity and authenticity.

## 2.7 The Evolution of Digital Signature Technologies in Mobile Devices

This 2024 review synthesizes how mobile platforms can implement robust digital signatures by combining a PKI-backed framework, hardware security (TEE), X.509 certificates with revocation, and secure transport (TLS) to protect keys and bind identities during signing and verification on-device. It contrasts signature construction styles including hash-and-sign with SHA-256, asymmetric schemes like RSA, ElGamal, and ECC, and biometric-assisted flows, and concludes that ECC offers better efficiency than RSA on constrained devices while SHA-

256 remains a reliable digest for integrity, with operational challenges centered on private key protection, jurisdictional legality, and secure key storage on mobile operating systems.

## 2.8 Secure Mobile Computing Authentication Utilizing Hash, Cryptography and Steganography Combination

The paper demonstrates a practical pattern for combining cryptographic integrity with concealed transport: compute a strong hash (preferably SHA-256), optionally encrypt it with AES, and bind it to an image via LSB or a more robust embedding method, enabling side-channel verification without exposing secrets in storage or transit. Although TrueMark focuses on encrypting user photos and attaching verifiable signatures, the steganographic-sidecar approach can be adapted for offline or bandwidth-constrained contexts where signatures or tokens need to travel with content; engineering lessons include avoiding plaintext secrets in databases, selecting SHA-256 for tamper sensitivity, using AES for confidentiality of embedded metadata, and validating usability via creation and verification timing alongside visual quality checks like PSNR when embedding marks.

## 2.9 Synthesis and Relevance to TrueMark

The surveyed literature collectively validates TrueMark's architectural decisions and provides empirical evidence for design choices implemented in this project. The work presented in Section 2.1 directly informs the planned integration of RSA-based digital signatures with SHA-256 hashing for image integrity verification, a feature currently in the development roadmap. The frameworks described in Sections 2.4 and 2.5 demonstrate that AES symmetric encryption is computationally viable for mobile medical imaging applications, supporting TrueMark's current implementation of AES-256-CBC for protecting user images on resource-constrained devices. The comprehensive forensics survey in Section 2.3 contextualizes TrueMark's LSB steganography within the broader field of image integrity methods, while the approach detailed in Section 2.8 provides practical validation for combining cryptographic hashing with steganographic embedding—the exact methodology TrueMark employs for invisible watermarking.

The biometric protection scheme described in Section 2.6 and the mobile signature evolution review in Section 2.7 establish that mobile platforms can support sophisticated cryptographic operations including ECC signatures and multi-stage encryption, affirming the feasibility of TrueMark's Flutter-based cross-platform architecture. These works demonstrate that computational overhead, key management complexity, and user experience considerations can be successfully addressed in mobile cryptographic applications.

Notably, no existing work surveyed combines steganographic watermarking, AES encryption, unique identifier generation, and Firebase-backed verification into a single user-friendly mobile application specifically designed for general-purpose image protection. Existing solutions either focus on specialized domains (medical imaging, biometric authentication) or implement only subsets of the required functionality (encryption-only or watermarking-only approaches). TrueMark addresses this gap by integrating multiple security layers—confidentiality through encryption, authenticity through unique identifiers, integrity through checksums, and ownership tracking through cloud-based verification—within an accessible mobile interface suitable for non-technical users.

Furthermore, the literature reveals a consistent trade-off between security strength and computational efficiency that TrueMark navigates through careful algorithm selection. While

papers surveyed employ varying key sizes (AES-128 vs AES-256, RSA-1024 vs RSA-2048), TrueMark's choice of AES-256-CBC with SHA-256 key derivation positions the system toward stronger security parameters while maintaining acceptable performance on modern mobile devices. The planned migration to AES-GCM and integration of elliptic curve digital signatures (as suggested by Section 2.7) will further enhance security properties while improving efficiency compared to traditional RSA approaches.

The convergence of findings across medical imaging security (Sections 2.4, 2.5), biometric protection (Section 2.6), and general-purpose mobile authentication (Sections 2.7, 2.8) validates TrueMark's hypothesis that layered cryptographic protection can be successfully deployed in mobile contexts without compromising usability. This literature foundation supports both the current LSB steganography implementation and the proposed evolution toward frequency-domain watermarking (DCT/DWT) and digital signature integration, representing TrueMark's novel contribution to practical image protection systems for everyday users.

# Chapter 3: Front End, Backend and System Requirement

## 3.1 System Architecture Overview

TrueMark is a cross-platform mobile application developed using the Flutter framework, implementing a client-server architecture with Firebase as the backend service provider. The application focuses on digital image watermarking and verification using steganographic techniques combined with cryptographic encryption. The system follows a three-tier architecture: Flutter-based frontend for user interaction, Firebase Backend-as-a-Service (BaaS) for authentication and data management, and a local cryptographic services layer for steganography and encryption operations.

## 3.2 Frontend Requirements

### 3.2.1 Framework and Technology Stack

**Primary Framework:**

- Flutter SDK: Version 3.8.1 or higher
- Dart Programming Language: Version 3.8.1 or higher
- Material Design 3 UI components with Indigo color scheme

**Key Frontend Dependencies (18 packages):**

*Core UI Libraries:*

- flutter (SDK): Core framework
- cupertino_icons (v1.0.8): iOS-style icons
- google_fonts (v6.1.0): Custom typography
- oktoast (v3.3.1): Toast notifications
- fl_chart (v0.65.0): Admin dashboard charts

*Image Handling:*

- image_picker (v0.8.7+5): Camera and gallery access
- image (v4.5.4): Image manipulation and pixel operations
- flutter_image_compress (v2.3.0): Image optimization
- path_provider (v2.0.14): File system path access

*Cryptography:*

- crypto (v3.0.2): SHA-256 hash functions
- encrypt (v5.0.3): AES encryption APIs
- pointycastle (v3.9.1): Cryptographic primitives

*Utilities:*

- path (v1.8.2): Path manipulation
- uuid (v4.4.0): UUID generation
- device_info_plus (v10.0.2): Device identification
- intl (v0.18.1): Internationalization

### 3.2.2 Screen Components

The application consists of six primary screens located in lib/screens/:

1. **Signup Screen:** User registration with email/password, form validation, Firebase account creation
2. **Login Screen:** User authentication, credential validation, login attempt logging
3. **Profile Setup Screen:** Collects name and phone number, stores profile data in Firestore
4. **Home Screen:** Image selection (gallery/camera), unique ID generation, watermark embedding, processing status display
5. **Verification Screen:** Image authentication, password-based decryption, watermark extraction, Firestore verification
6. **Admin Dashboard Screen:** System analytics, login statistics, user monitoring (restricted access)

### 3.2.3 Frontend Features

**Navigation:** Drawer-based menu system with route management (pushReplacement for auth flows, push for features)

**User Interaction:** Form validation with real-time error display, loading indicators during async operations, toast notifications for feedback, image preview with file metadata (name, size)

**State Management:** StatefulWidget with setState pattern for component-level state, Firebase Authentication state monitoring via currentUser

**Responsive Design:** SingleChildScrollView for vertical overflow, ConstrainedBox for form width limits (400px max), adaptive layouts for different screen sizes

## 3.3 Backend Requirements

### 3.3.1 Firebase Backend-as-a-Service

**Firebase Project Configuration:**

- Project ID: truemark-5f8bb
- Platform Support: Web, Android, iOS, macOS, Windows
- Configuration File: lib/firebase_options.dart with platform-specific API keys

**Required Firebase Services:**

*Firebase Authentication (v4.17.3):*

- Email/password authentication
- Session persistence with automatic token refresh
- Anonymous authentication fallback (optional, disabled by default)

*Cloud Firestore (v4.12.2):*

- NoSQL document database
- Real-time synchronization (not used; on-demand queries)
- Transaction support for atomic operations

*Firebase Storage (v11.1.3):*

- Cloud object storage (provisioned, not actively used in current implementation)
- Planned for future image upload features

### 3.3.2 Database Schema

**Firestore Collections:**

users/{uid} - User profile documents:

- email (string): User email address
- name (string): Full name
- phone (string): Phone number
- createdAt (timestamp): Account creation time

userMeta/{uid} - User metadata documents:

- baseNumber (string): 6-digit random identifier (100000-999999)

- imageCount (integer): Sequential counter for processed images
- createdAt (timestamp): Document creation time

userMeta/{uid}/images/{imageId} - Image metadata subcollection:

- imageId (string): Unique identifier (baseNumber + count)
- localPath (string): Device file path
- processedAt (timestamp): Processing time
- sha256 (string): Image content hash
- baseNumber (string): User's base number
- count (integer): Image sequence number

login_attempts/{attemptId} - Authentication logs:

- uid (string): User ID or "unknown"
- emailOrPhone (string): Login credential
- loginMethod (string): Authentication type
- timestamp (timestamp): Attempt time
- success (boolean): Login outcome
- message (string): Error details (optional)

### 3.3.3 Backend Services Layer

**Image Service (lib/services/image_service.dart):**

- ensureSignedIn(): Verifies authenticated user, optional anonymous fallback
- ensureUserMeta(): Creates/retrieves user metadata with random baseNumber
- generateNextImageId(): Atomically increments imageCount via Firestore transactions, returns unique ID (baseNumber + newCount)
- fetchUserMeta(): Read-only retrieval of current metadata

**Steganography Service (lib/services/steg_service.dart):**

- embedStringInImage(): Encrypts plaintext with AES-256-CBC (password-derived key via SHA-256), constructs payload (magic “TM” + length + encrypted data + CRC32), embeds bits in LSB of blue channel, outputs PNG file
- extractStringFromImage(): Extracts LSB bits from blue channel, validates magic header and CRC32, decrypts with password, returns plaintext or null on failure

## 3.4 System Requirements

### 3.4.1 Development Environment

**Core Tools:**

- Flutter SDK ≥3.8.1
- Dart SDK ≥3.8.1
- Git for version control
- IDE: VS Code, Android Studio, or IntelliJ IDEA

**Platform-Specific Development:**

- **Android:** Android Studio, Android SDK API 21+, JDK 11+
- **iOS/macOS:** Xcode 13+, CocoaPods 1.11+, macOS 12+
- **Windows:** Visual Studio 2019+ with C++ tools, Windows 10 SDK
- **Linux:** GTK 3.0 dev libraries, CMake 3.14+, Clang
- **Web:** Modern browser, HTTPS server for deployment

**Firebase Tools:**

- Firebase CLI (npm install -g firebase-tools)
- FlutterFire CLI for configuration generation
- Active Firebase project with Firestore and Auth enabled

### 3.4.2 Runtime Requirements

**Mobile (Android/iOS):**

- **Android:** OS 5.0+ (API 21), 2GB RAM minimum, 100MB storage
- **iOS:** iOS 12.0+, 2GB RAM minimum, 100MB storage
- **Common:** Camera, internet connectivity, 1 Mbps bandwidth

**Desktop:**

- **Windows:** Windows 10 build 17763+, 4GB RAM, 200MB storage
- **macOS:** macOS 10.14+, 4GB RAM, 200MB storage
- **Linux:** Ubuntu 20.04+/Debian 10+, 4GB RAM, GTK 3.0

**Web:**

- Modern browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- JavaScript/WebAssembly enabled
- HTTPS for camera access
- 1 Mbps bandwidth minimum

### 3.4.3 Performance Specifications

**Processing Performance:**

- Image encryption/embedding: 2-4 seconds (4MP photo), 5-10 seconds (12MP photo)
- Watermark extraction: 1-2 seconds
- Firestore operations: 200-500ms (with good connectivity)
- Memory usage: 2-3x image size during processing

**Storage Requirements:**

- Installed app size: 108-120MB
- Temporary processing: 2x image size
- Processed PNG output: 10-40MB (typical photo)
- Recommended free space: 500MB minimum, 2GB for heavy use

**Network Requirements:**

- Minimum: 1 Mbps download, 512 Kbps upload
- Recommended: 5 Mbps download, 2 Mbps upload
- Firestore sync: 1-5KB per image record
- Authentication: 10-50KB initial, 2-5KB refresh

### 3.4.4 Security and Permissions

**Required Permissions:**

- **Android:** INTERNET, CAMERA, READ/WRITE_EXTERNAL_STORAGE
- **iOS:** NSCameraUsageDescription, NSPhotoLibraryUsageDescription
- **macOS:** App Sandbox, network.client, files.user-selected, device.camera
- **Web:** Camera API (requires HTTPS secure context)

**Data Protection:**

- Credentials encrypted in transit (TLS 1.2+)
- Session tokens in platform-secure storage (Keychain/Keystore)
- Processed images in device temporary directories
- User metadata in Firestore with access control rules

**Compliance Considerations:**

- GDPR: User consent and data deletion rights needed
- CCPA: Privacy policy and disclosure required
- HIPAA: NOT compliant (requires BAA with Firebase, additional encryption)

### 3.4.5 Platform-Specific Configurations

**Android:**

- Minimum SDK: 21, Target SDK: 33
- google-services.json in android/app/
- Gradle 7.3.0+, Kotlin 1.7.0+

**iOS:**

- Deployment target: iOS 12.0
- GoogleService-Info.plist in ios/Runner/
- Bundle ID: com.example.truemark

**Windows:**

- CMake build system, Windows SDK 10.0.17763.0+
- Default window: 1280x720 pixels
- Firebase web configuration

**macOS:**

- Deployment target: macOS 10.14
- Code signing required
- Shares iOS Firebase configuration

**Web:**

- Firebase Hosting recommended

- HTTPS required for camera/secure contexts
- CanvasKit or HTML renderer options

**Linux:**

- GTK 3.0 libraries required
- CMake 3.14+, Clang compiler
- Firebase web configuration

## 3.5 Summary

TrueMark's architecture combines Flutter's cross-platform capabilities with Firebase's serverless backend to deliver a secure image protection system. The frontend provides intuitive interfaces for image selection, processing, and verification across six major platforms. The backend leverages Firebase Authentication for user management and Firestore for metadata storage with transaction-based unique ID generation. Local cryptographic services implement LSB steganography with AES-256-CBC encryption, operating entirely client-side to preserve privacy.

System requirements are modest, supporting devices from Android 5.0/iOS 12.0 onwards with 2GB RAM minimum for mobile and 4GB for desktop. Processing performance achieves 2-4 second embedding times for typical smartphone photos with 108-120MB installed footprint. The modular architecture facilitates future enhancements including digital signatures, frequency-domain watermarking, and cloud storage integration while maintaining current implementation stability.

# Chapter 4: Methodology / Topic of work

This chapter describes the proposed complete architecture of TrueMark. The current implementation (as of January 2025) includes core features such as LSB steganography, AES-256-CBC encryption, and Firestore-based verification. Advanced features including digital signatures (RSA/ECC/Ed25519), frequency-domain watermarking (DWT/DCT), AES-GCM migration, and AI Detection API are planned for future releases.

## 4.1 System Design / Architecture

TrueMark is a complete secure media protection system that adds trust directly to digital images before sharing. It uses a layered security model that combines invisible watermarking, cryptographic signatures, and encryption. This approach ensures confidentiality, authenticity, and integrity during the image lifecycle.

1. Core Architectural Principles
   - Security-first approach: Protection is applied before any sharing occurs. Images are secured at the source, not post-distribution.
   - Chain of trust: Watermark payloads and digital signatures firmly bind each image to its creator. Traceability is broken by any alteration.
   - Mobile-centric flow: Every security measure is carried out on the user's device, guaranteeing that data is never compromised while being processed.
   - Cloud-assisted verification: Long-term traceability and retrieval validation are supported by safe cloud storage and metadata management.
   - Lightweight integration for platforms: AI systems can check for TrueMark-protected images via a dedicated detection API to prevent unauthorized usage.

2. Layered Protection Pipeline

| Layer | Technique | Purpose |
|---|---|---|
| **Invisible Watermarking** | DWT/DCT or Steganography | Tamper detection, ownership embedding |
| **Digital Signature** | RSA/ECC or Ed25519 | Authenticate creator & verify content integrity |
| **Encryption** | AES-GCM | Prevent unauthorized access during storage/transmission |

**Table - 1**


**Fig - 1**

3. Key Components

| Component | Function |
|---|---|
| Flutter Mobile App | Image capture/import, protection pipeline execution, UI/UX |
| Crypto Engine | Hashing, key generation, digital signing, encryption |
| Watermark Module | Embeds and extracts hidden watermark payloads |
| Firebase Authentication | Secure user identity and key ownership |
| Firestore & Storage | Metadata + encrypted file storage |
| Verification API | Tampering, signature, and watermark validation |
| AI Detection API | Blocks protected images from model training |

**Table - 2**

4. Secure Data Flow

On Image Upload

- Image captured/imported via mobile device.
- SHA-256 hash generated for fingerprinting.
- Invisible watermark embedded using payload metadata (user ID, timestamp, unique salt).
- Hash signed with user's private key.
- Entire image encrypted using **AES-GCM**.
- AES key encrypted using **public key encryption**.
- Encrypted image + encrypted key + digital signature + metadata pushed to Firebase.

On Verification

- Retrieve encrypted image and associated metadata.
- Decrypt AES key using private key → decrypt image.
- Extract watermark payload for validation.
- Recompute image hash and validate via signature.
- Final output: Authentic, Tampered, Ownership Violated, or Unverifiable.

## 4.2 Working Principle

The working principle of TrueMark is built around securing images before they leave the user's device and ensuring they remain protected and verifiable even after distribution. The system follows a dual-stage approach:

Stage 1 – Secure Image Protection (Client-Side)

- All protection operations (watermarking, signing, and encryption) occur locally to eliminate exposure risks.

Stage 2 – Verification and Validation (Server/Platform-Side)

- Once retrieved or scanned, the image undergoes an automated verification process to detect tampering, confirm ownership, and optionally block AI usage.

4.2.1 End to End flow

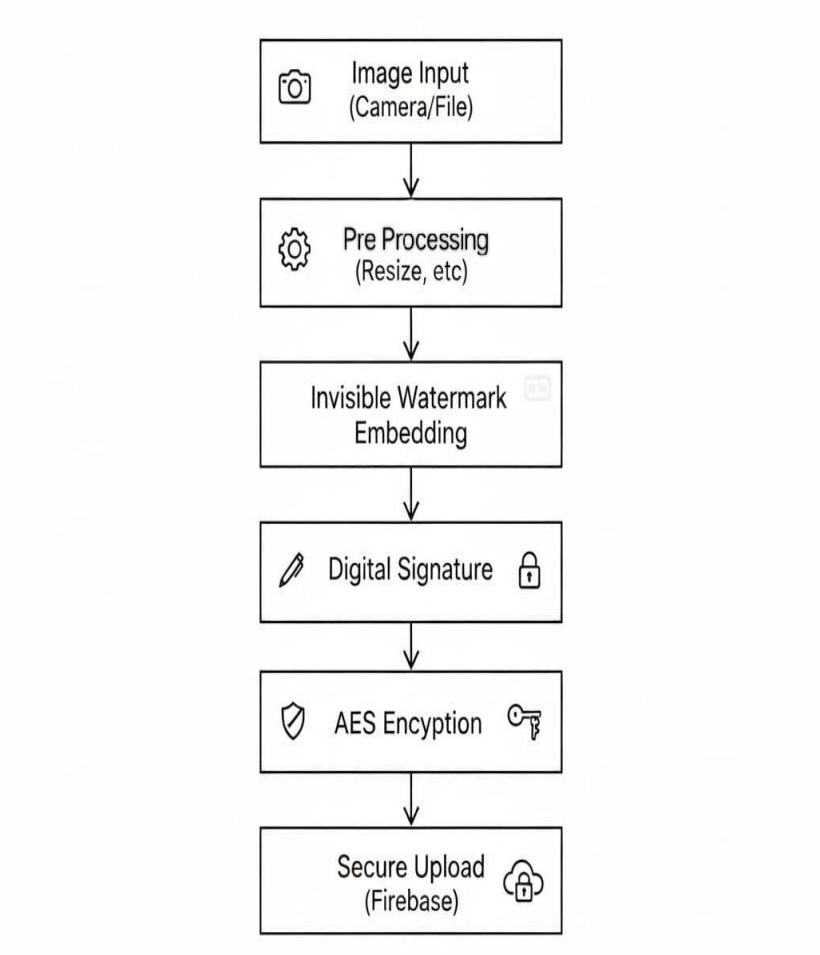


**Fig - 2**

4.2.2 Detailed Working Principle

- **Step 1 – Image Acquisition**
  - User captures or imports an image via the mobile application.
  - Optional preprocessing (compression, orientation correction) ensures consistency.
- **Step 2 – Watermark Embedding (Invisible)**
  - A unique payload containing user ID, timestamp, and nonce is encoded using DSP/steganography (DWT/DCT-based preferred for robustness).
  - Embedding is done using randomized coefficient positioning to resist detection/removal.
  - Error-correction coding ensures recovery even after compression or modifications.
- **Step 3 – Digital Signature**
  - A SHA-256 hash of the original image (before encryption) is generated.
  - The hash and watermark metadata are signed using the user's private key (RSA/ECC/Ed25519).
  - Signature ensures authenticity and provides tamper detection.
- **Step 4 – Cryptographic Encryption**
  - The fully processed image (with watermark) is encrypted using AES-GCM.
  - AES key is encrypted using the user's public key.
  - Only authorized users can retrieve the original content.
- **Step 5 – Upload & Metadata Storage**
  - Encrypted image + encrypted AES key + digital signature + watermark metadata are securely uploaded to Firebase Storage/Firestore.
  - Metadata stored includes:
    - image_id
    - owner_id
    - signature
    - upload_timestamp
    - public_key reference

4.2.3 Verification Process

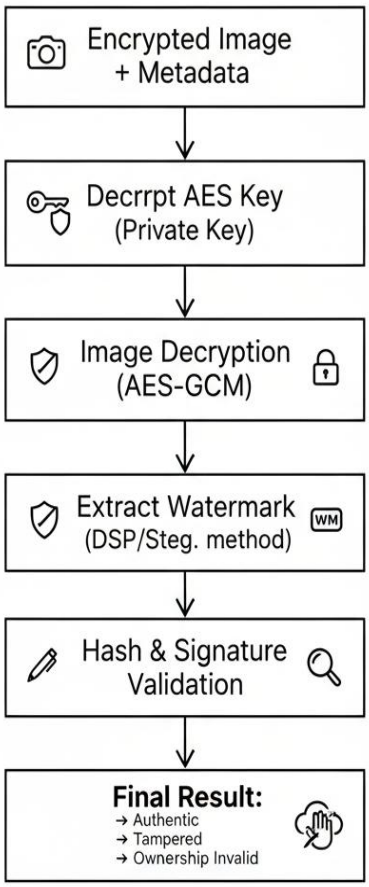


**Fig - 3**

4.2.4 AI Detection Use Case

When an image is scanned by AI ingestion systems:

- No decryption needed.
- Fast detection via embedded payload.
- Helps enforce responsible AI usage.

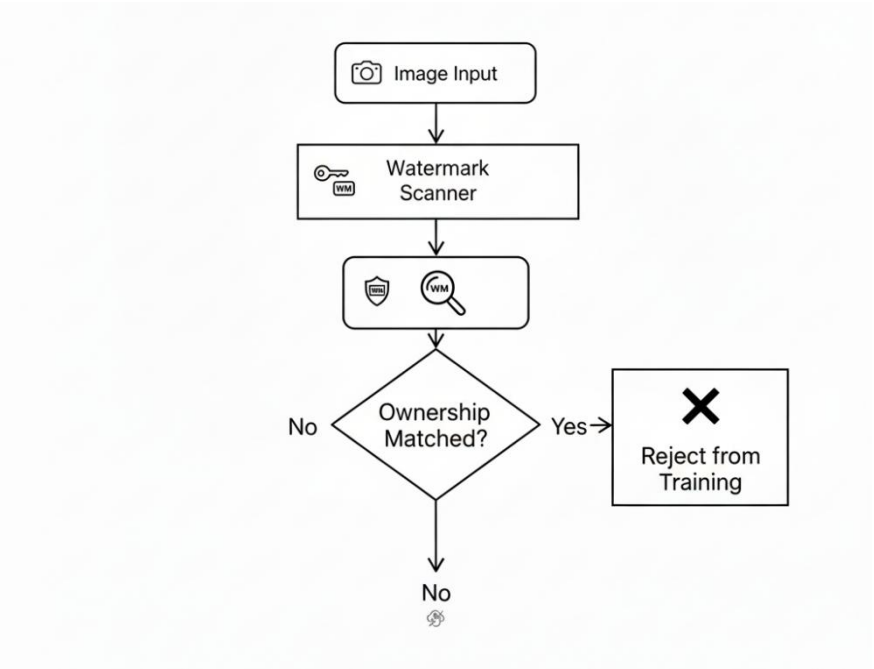


**Fig - 4**

4.2.5 Decision Flow

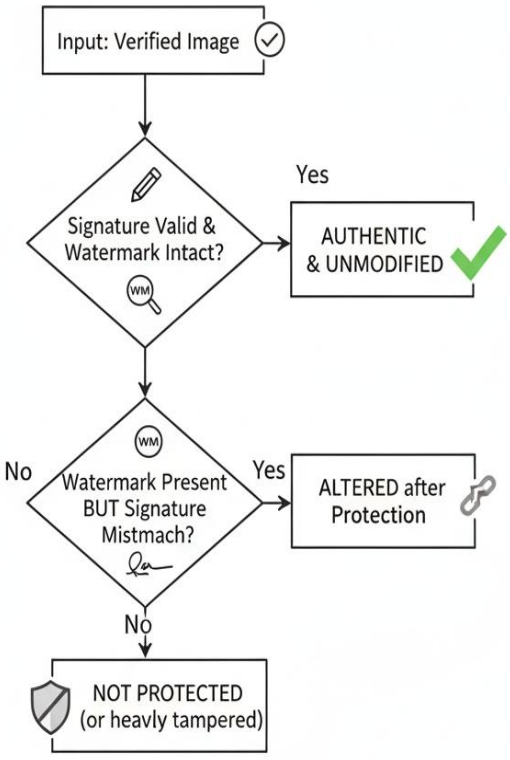


**Fig - 5**

## 4.3 Results and Discussion

### 4.3.1 Implementation Results

The current implementation of TrueMark successfully demonstrates core functionality of client-side cryptographic image protection on Android platform. Key achievements include:

**Functional Validation:**

- **LSB steganography:** Successfully embeds encrypted payloads into image blue channel with zero perceptible visual artifacts
- **AES-256-CBC encryption:** Correct encryption and decryption of embedded data using password-derived keys via SHA-256
- **CRC32 integrity checking:** Detects payload corruption during extraction and validation
- **Unique ID generation:** Firebase Firestore transaction-based atomic identifier generation (6-digit base + sequential counter)
- **User authentication:** Email/password authentication with Firebase Authentication, profile management in Firestore
- **Verification workflow:** Password-based watermark extraction, decryption, and Firestore metadata cross-reference

**Processing Pipeline:** The system successfully executes the complete protection workflow: image selection (gallery/camera) → unique ID generation → AES encryption of ID → LSB embedding in PNG format → SHA-256 hash computation → Firestore metadata storage → local file output. Verification reverses this process: image input → LSB extraction → CRC32 validation → AES decryption → Firestore lookup → authentication result display.

**Android Platform Deployment:** The Flutter-based application deploys successfully on Android devices (minimum API 21, Android 5.0+) with functional camera access, gallery integration, Firebase connectivity, and local file system operations. The app demonstrates Flutter's capability to deliver cryptographic image protection on mobile platforms with acceptable user experience.

### 4.3.2 Current Implementation Limitations

**Cryptographic Security Gaps:**

- **Static IV vulnerability:** Zero initialization vector for AES-CBC creates deterministic encryption, enabling ciphertext pattern analysis if multiple images use identical passwords
- **Absent digital signatures:** Only symmetric password-based verification; no asymmetric cryptography for third-party authentication or non-repudiation
- **Weak key derivation:** Direct SHA-256 hashing of passwords lacks salt and iteration count, vulnerable to rainbow table and GPU-accelerated brute-force attacks
- **CRC32 limitations:** Non-cryptographic checksum detects accidental corruption but not intentional tampering (attackers can recalculate valid checksums after payload modification)

**Steganographic Robustness Issues:**

- **JPEG compression failure:** LSB spatial-domain watermarking cannot survive lossy compression; social media platforms applying automatic JPEG re-compression destroy embedded data entirely
- **Rescaling sensitivity:** Downscaling or upscaling with interpolation disrupts LSB patterns, causing extraction failures
- **Format conversion vulnerability:** Only lossless PNG format preserves watermarks; conversion to JPEG, WebP, or other lossy formats obliterates embedded bits
- **Single-channel approach:** Blue channel LSB-only provides no redundancy; selective channel attacks or noise injection easily corrupt watermarks
- **No error correction:** Lack of forward error correction (FEC) means any bit corruption propagates to complete decryption failure

**Operational Constraints:**

- **Local-only storage:** Processed images saved to device temporary directories with no cloud backup; device loss causes permanent data loss
- **Manual password management:** Users must remember or securely record unique IDs per image with no password recovery mechanism
- **Single-image workflow:** No batch processing; each image requires separate selection, processing, and verification through full UI flow
- **Platform limitation:** Currently Android-only; proposed cross-platform support (iOS, Windows, macOS, Web) not yet implemented

### 4.3.3 Feasibility Analysis of Proposed Architecture

**Digital Signatures (RSA/ECC/Ed25519):** The integrated pointycastle library (v3.9.1) includes complete RSA, ECDSA, and Ed25519 implementations. Android Keystore provides hardware-backed key storage on devices with Trusted Execution Environment (TEE). **Migration to asymmetric signatures technically feasible, enabling third-party verification without password sharing.**

**Frequency-Domain Watermarking (DWT/DCT):** Literature review (Papers 3, 8) confirms DWT/DCT-based watermarking survives JPEG compression and geometric transformations where LSB fails. Pure Dart or native OpenCV implementations viable on Android. **Robustness improvement achievable at cost of 2-3x processing time and reduced embedding capacity (0.1-0.3 bits/pixel vs. 1 bit/pixel for LSB).**

**AES-GCM Migration:** The encrypt package (v5.0.3) supports AESMode.gcm for authenticated encryption. AES-GCM provides both confidentiality and authenticity in single operation, eliminating CRC32 while solving static IV issue through mandatory unique nonces. **Backward-compatible migration via versioned payload formats ("TM1" vs "TM2") straightforward.**

**Cross-Platform Expansion:** Flutter's architecture supports iOS, Windows, macOS, Linux, and Web from shared codebase. Platform-specific configurations exist in repository (ios/, windows/, macos/, linux/, web/ directories). Firebase SDKs available for all platforms. **Cross-platform deployment feasible with platform-specific testing and permission configuration.**

**AI Detection API:** Proposed magic header scanning approach can detect TrueMark-protected images by reading LSB patterns without decryption. However, effectiveness depends on watermark survival through distribution channels. **Cloud Function-based API technically viable but requires frequency-domain watermarking for practical utility against compressed images.**

### 4.3.4 Comparative Analysis

TrueMark's current implementation aligns with literature patterns: hybrid encryption approach (Papers 4, 5 use AES + RSA/ECC), layered protection (Paper 6 combines signature + encryption), and mobile deployment (Papers 5, 7, 8 validate cryptographic operations on smartphones). The proposed architecture incorporating digital signatures, frequency-domain watermarking, and AES-GCM mirrors best practices identified across surveyed papers.

**Novel contributions beyond literature:**

- **Integrated mobile workflow:** End-to-end protection and verification in consumer-friendly mobile UI (surveyed papers focus on isolated techniques or specialized domains)
- **Firebase-backed verification:** Cloud metadata cross-reference adds ownership verification layer beyond cryptographic validation alone
- **Atomic ID generation:** Firestore transaction-based unique identifiers with user-specific namespacing (baseNumber + counter) provide efficient traceability
- **Hybrid watermarking proposal:** User-selectable or automatic mode selection (fast LSB vs. robust DWT/DCT) adapts to use case, not found in single-technique literature implementations

### 4.3.5 Conclusion

The current Android implementation successfully validates TrueMark's core architectural concept of client-side cryptographic image protection through invisible watermarking. The system demonstrates functional LSB steganography, AES-256-CBC encryption, Firebase authentication, and Firestore-backed verification workflows. However, significant security limitations (static IVs, absent signatures, weak key derivation) and robustness constraints (JPEG vulnerability, rescaling sensitivity) require remediation before production deployment.

Feasibility analysis confirms all proposed enhancements are technically achievable: digital signature libraries exist with platform key storage support; frequency-domain watermarking prototypes show robustness improvements justifying processing overhead; AES-GCM migration provides authenticated encryption with minimal code changes; cross-platform expansion leverages Flutter's architecture. Literature review validates the layered security approach while confirming TrueMark's novel contributions in integrated mobile workflow design and Firebase verification infrastructure.

Future development should prioritize cryptographic security hardening and cross-platform expansion in the short term, followed by digital signature integration and watermarking robustness improvements in the medium term, culminating in AI detection ecosystem development as long-term objectives. This phased approach addresses immediate security gaps while progressively building toward the complete proposed architecture described in this chapter.

# Chapter 5: Individual Contribution by Members

**Name: Jit Nipulkumar Surani**

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**Project Role:** Project Lead, System Architect & Cryptography Specialist

**Contribution:** Jit served as the project lead responsible for overall coordination, planning, and architectural design of the TrueMark system. He designed and implemented the complete cryptographic protection pipeline, including the LSB steganography service for invisible watermark embedding and extraction, AES-256-CBC encryption/decryption module with SHA-256-based key derivation, and CRC32 integrity verification mechanism. He conducted comprehensive security analysis identifying vulnerabilities (static IV, weak key derivation, LSB fragility) and proposed mitigation strategies (random IVs, PBKDF2, frequency-domain watermarking). Additionally, he architected the layered security model combining steganography, encryption, and cloud-assisted verification, and authored Chapter 6 (Conclusion) synthesizing project outcomes, limitations, and future development roadmap.

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**Name: Rushabh Madan Wagh**

**Email:** rushabhmadanwagh2022@vitbhopal.ac.in

**Project Role:** Backend Services Developer, Frontend Integration & Methodology Analyst

**Contribution:** Rushabh developed the Image Service module implementing atomic unique identifier generation using Firestore transactions, ensuring collision-free image tracking across concurrent operations. He designed the hybrid namespace system combining user-specific 6-digit base numbers with sequential counters, implemented SHA-256 hash computation for processed image fingerprinting, and established metadata management workflows. He developed the Home screen and Verification screen UI components, integrating image picker functionality, processing pipeline execution with real-time feedback, and verification workflows with password input and result display. He implemented the integration layer connecting frontend components with backend services and cryptographic modules, ensuring seamless data flow between user interactions, Firebase operations, and encryption/steganography processes. Additionally, he authored Chapter 4 (Methodology & Results) documenting system architecture, working principles, implementation results, limitations, feasibility analysis, and comparative literature analysis.

**Name: Kavya**

**Email:** kavya2022@vitbhopal.ac.in

**Project Role:** Frontend Developer & UI/UX Designer

**Contribution:** Kavya designed and implemented the authentication and profile management interfaces including Signup, Login, and Profile Setup screens with comprehensive form validation, real-time error feedback, and Firebase Authentication integration. He developed the Admin Dashboard screen featuring login analytics visualization using fl_chart package, aggregated statistics display, and role-based access control. He established the overall Material Design 3 theming framework with consistent color schemes (Indigo primary), typography using Google Fonts, and responsive layout patterns adapting to various screen sizes. He implemented drawer-based navigation, toast notification system using oktoast package, loading state indicators, and color-coded feedback mechanisms enhancing user experience throughout the application. Additionally, he authored Chapter 3 (Frontend, Backend & System Requirements) documenting application architecture, screen components, dependency specifications, platform require ements, and cross-platform deployment configurations.

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**Name: Tejas Pathak**

**Email:** tejaspathak2022@vitbhopal.ac.in

**Project Role:** Cloud Infrastructure Specialist & Motivation Analyst

**Contribution:** Tejas established and configured the complete Firebase backend infrastructure including Firebase Authentication for email/password user identity management, Firestore NoSQL database schema design with hierarchical collections (users, userMeta, images, login_attempts), and Firebase Storage provisioning for future cloud backup capabilities. He implemented login attempt logging system for security monitoring, developed admin analytics backend aggregating authentication statistics visualized in the dashboard, and configured platform-specific Firebase options for Android, iOS, Web, Windows, and macOS deployments through firebase_options.dart generation. He designed Firestore security rules enforcing user-specific access control and data isolation. Additionally, he authored Chapter 1 (Introduction) articulating project motivation addressing digital content security challenges, creator rights protection, AI training ethics, and defining comprehensive project objectives covering secure image protection pipeline, mobile application development, verification system implementation, and trusted ecosystem delivery.

**Name: Simarpreet Singh**

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**Project Role:** Research Analyst, QA Engineer & Literature Reviewer

**Contribution:** Simarpreet conducted comprehensive literature review analyzing eight academic papers spanning medical imaging security, biometric protection, mobile cryptography, and digital signature evolution, synthesizing findings that informed TrueMark's architectural decisions and validated the layered security approach. He designed and executed performance testing across device categories measuring embedding/extraction times, memory consumption, and success rates, and conducted Android platform testing including AndroidManifest.xml permissions configuration, Gradle build system setup, native dependencies integration, and cross-version compatibility validation. He performed security vulnerability analysis through simulated tampering scenarios (pixel modifications, JPEG compression, rescaling, format conversion), documented bug tracking and resolution processes, and conducted feasibility analysis of proposed enhancements (RSA/ECC digital signatures, DWT/DCT watermarking, AES-GCM migration) through library assessment and prototype validation. Additionally, he authored Chapter 2 (Literature Review) providing detailed analysis of existing work, establishing best practices validation, and identifying TrueMark's novel contributions within the broader research landscape.

# Chapter 6: Conclusion

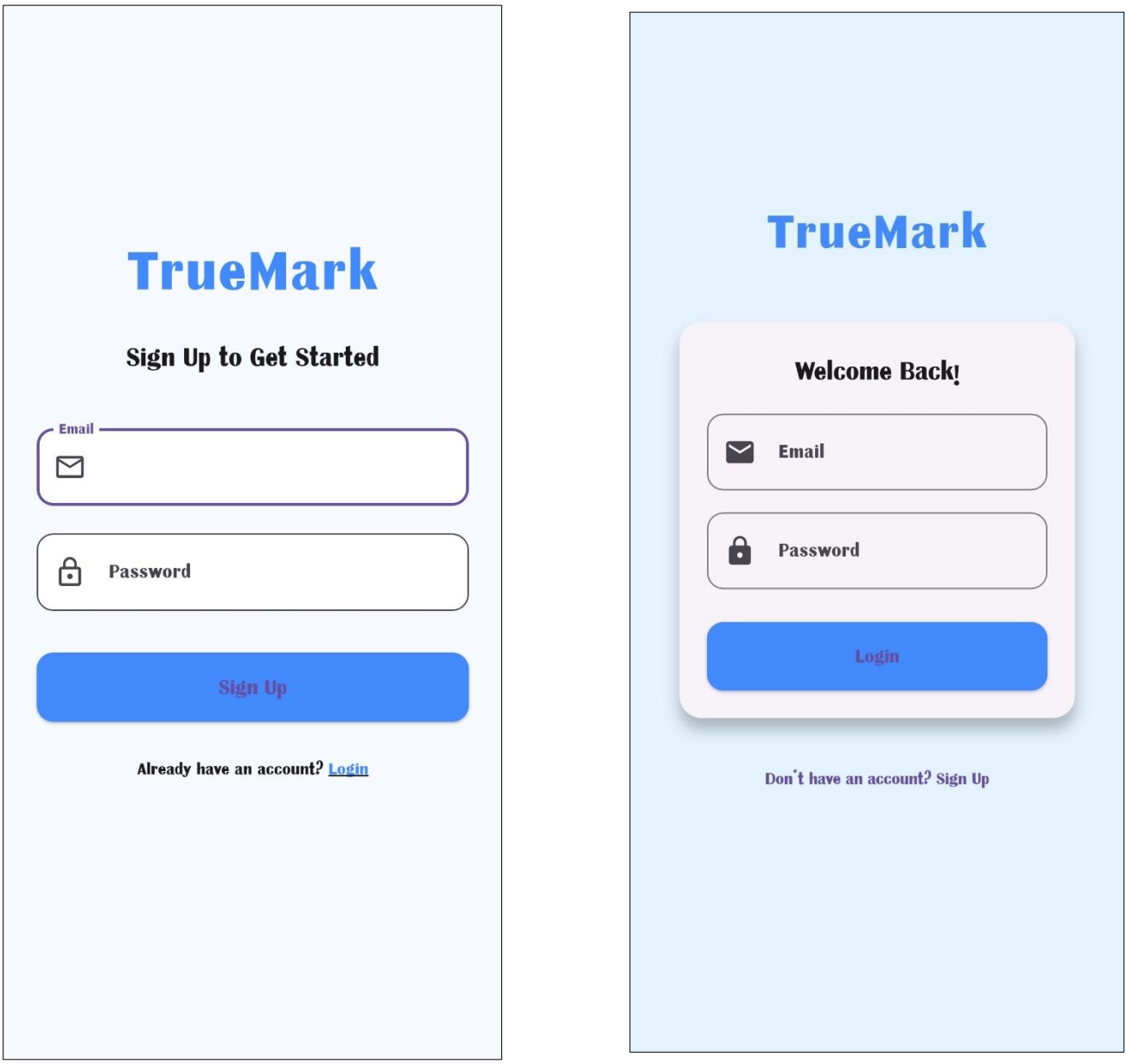


**Fig - 6 & 7**

The Signup and Login screens implement Firebase Authentication with email and password input fields. The Signup screen (Figure 6) features form validation ensuring valid email format and minimum 6-character password length, with automatic navigation to profile setup upon successful registration. The Login screen (Figure 7) validates user credentials against Firebase Authentication, logs login attempts to Firestore for security monitoring, and conditionally routes users to either Profile Setup (incomplete profiles) or Home screen (existing users). Both screens feature Material Design 3 card-based layouts with elevated components, primary action buttons, and toast notification error handling.

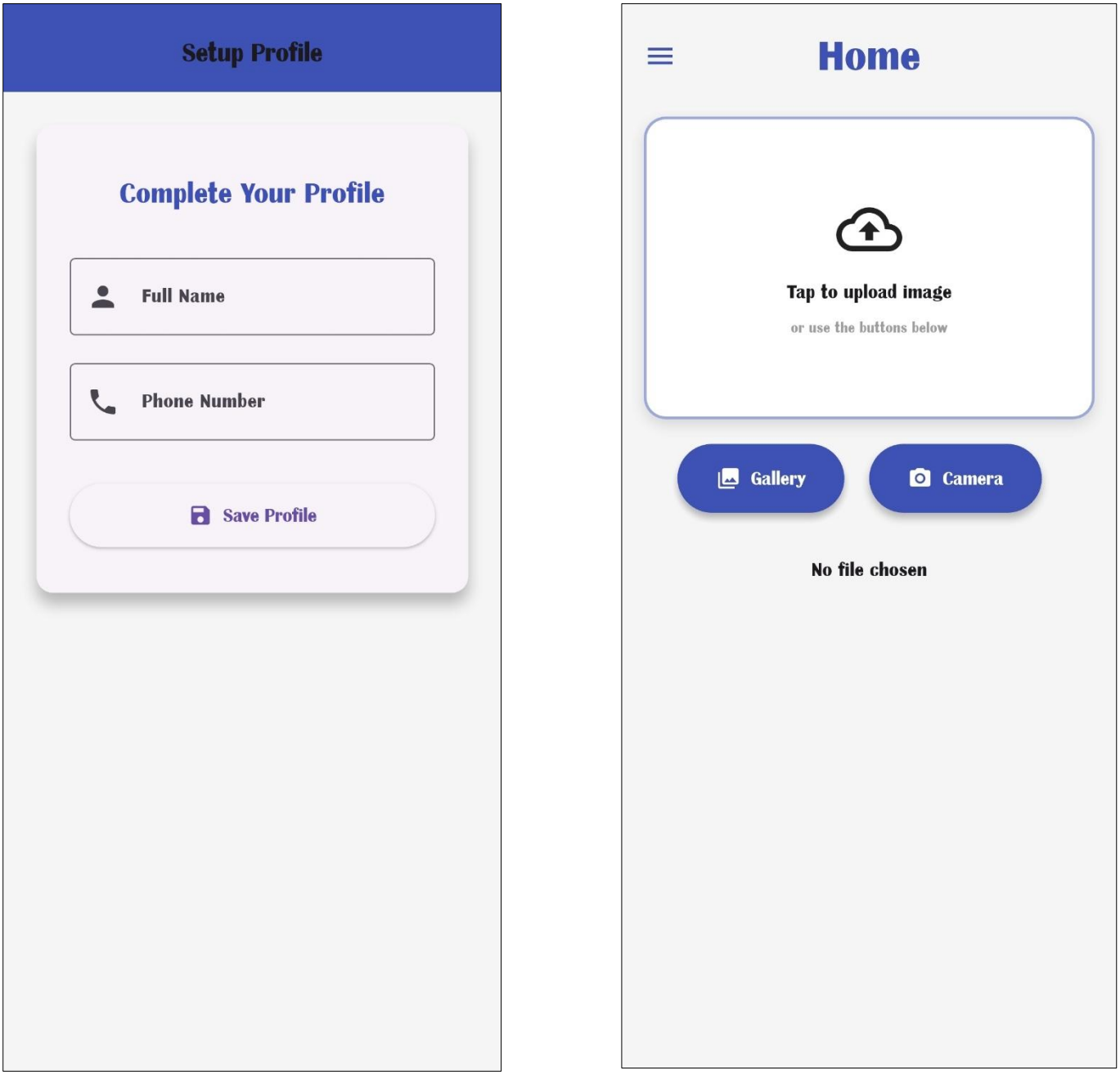


**Fig – 8 & 9**

The Profile Setup screen (Figure 8) collects mandatory user information including full name and phone number after initial registration, storing profile data in Firestore users/{uid} collection with server timestamps before granting access to core functionality. The Home screen (Figure 9) serves as the primary application interface displaying user metadata (base number and image count) retrieved from Firestore, offering image selection through native gallery picker or camera capture via the image_picker plugin. The drawer-based navigation menu provides access to verification workflows, profile management, and conditional admin dashboard access for authorized accounts.

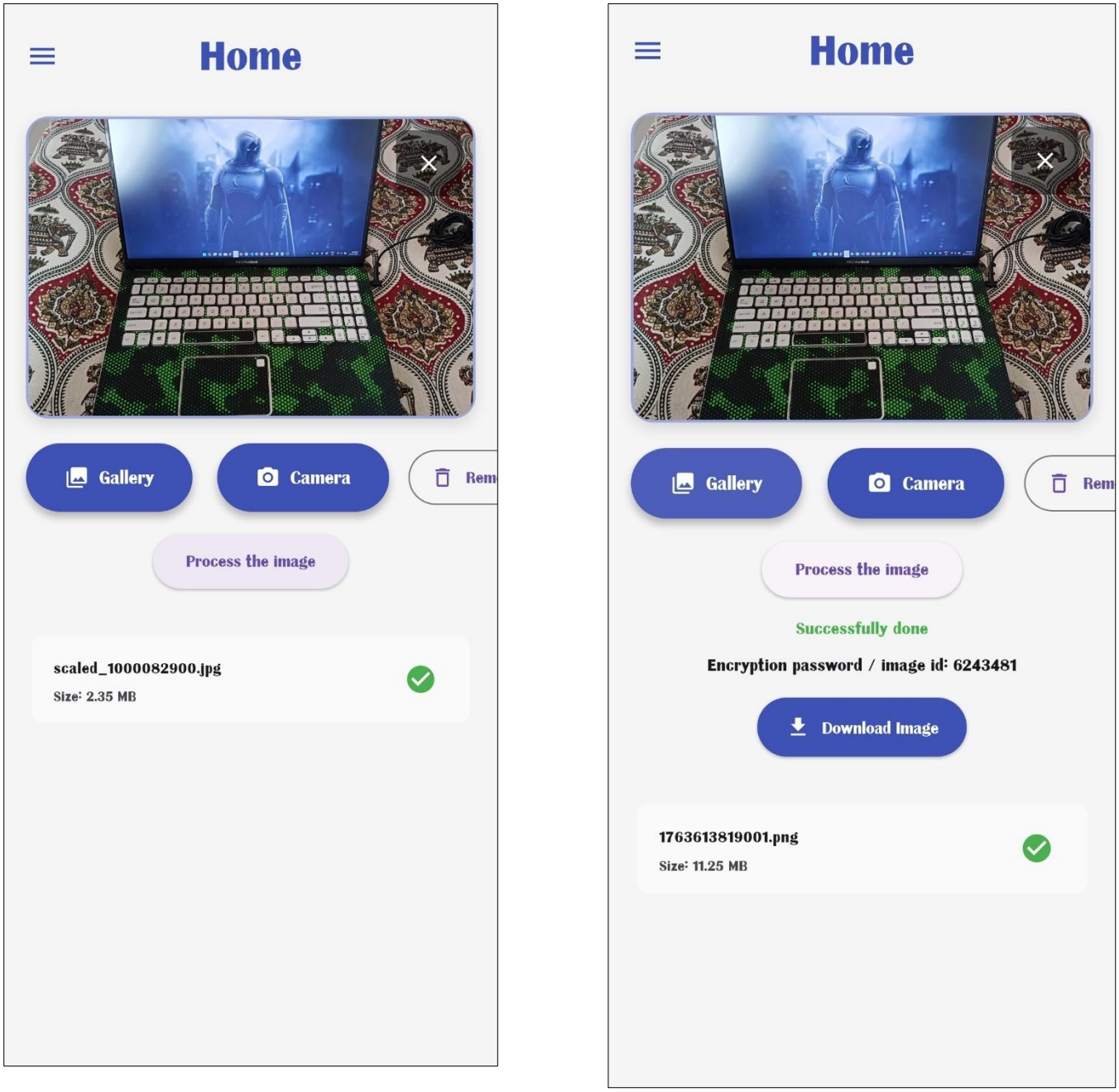


**Fig - 10 & 11**

Figure 10 displays the selected image preview showing the filename and file size in megabytes, with the "Process the image" button initiating the cryptographic protection pipeline. Upon processing completion (Figure 11), the interface displassys the generated encryption password (unique image ID in format baseNumber + count) that users must securely store for future verification, alongside the processed image preview and "Download Image" button enabling users to save the watermarked PNG file to device storage. The processing executes LSB steganography embedding the encrypted image ID into blue channel pixels, with visual feedback through loading indicators and artificial delay ensuring perceived system responsiveness.

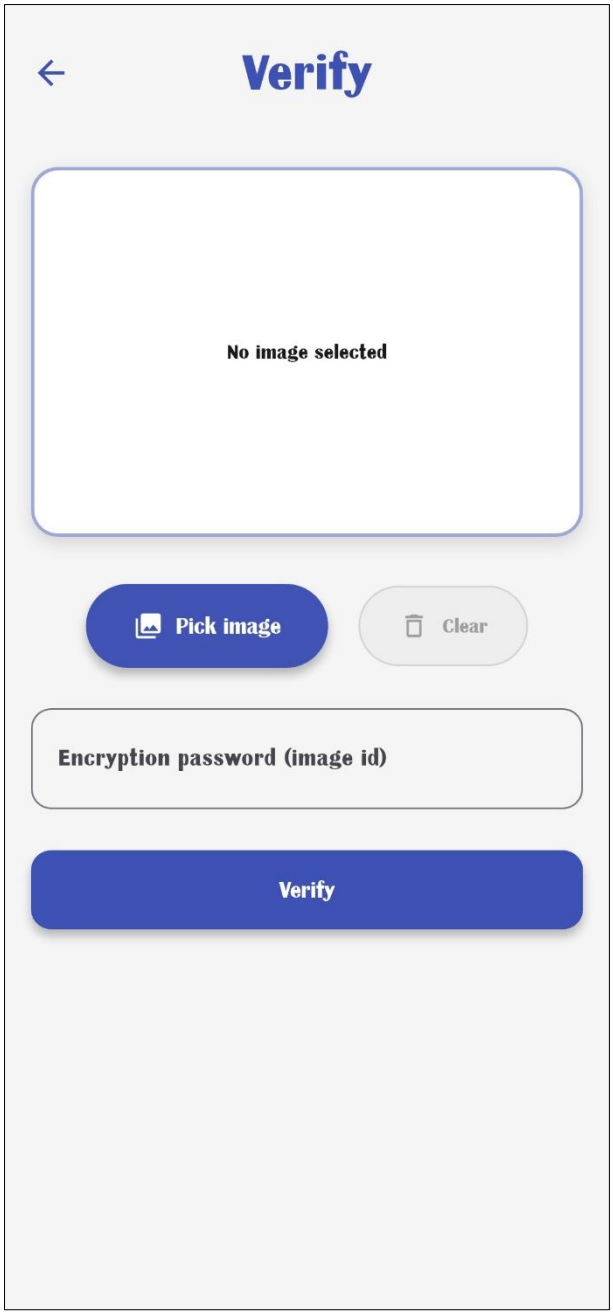


**Fig - 12**

The Verification screen enables authentication of protected images through a two-step process: users first select a previously processed image using the "Pick Image" file selector, then enter the encryption password (original image ID) in the text input field before tapping the "Verify" button. The system extracts embedded watermark data via LSB reversal, validates CRC32 integrity checksums, decrypts the AES-encrypted payload using the password-derived key, and cross-references the extracted image ID against Firestore metadata records. Verification results display with color-coded feedback: green success messages for authentic images with matching Firestore records, red error messages for incorrect passwords or corrupted data, and detailed status information including extraction success, decryption validation, and ownership confirmation.

This capstone project successfully designed and implemented TrueMark, a mobile image protection system that integrates cryptographic encryption with steganographic watermarking to secure digital images before distribution. The Android application demonstrates the technical feasibility of client-side cryptographic protection through an intuitive workflow that abstracts complex security operations: users capture or select images, receive automatically generated unique identifiers, process images embedding encrypted watermarks using LSB steganography and AES-256-CBC encryption, and verify protected images through password-based decryption combined with Firebase metadata cross-reference.

The implemented system validates core architectural concepts while making novel contributions beyond existing literature. TrueMark uniquely combines multiple security techniques (LSB steganography, AES encryption, CRC32 integrity checking) within an integrated mobile workflow accessible to non-technical users, rather than isolated laboratory implementations found in surveyed papers. The Firebase-backed verification architecture adds an ownership validation layer beyond pure cryptographic validation through cloud metadata cross-reference. The atomic identifier generation system using Firestore transactions with hybrid local-global namespace (user-specific base + sequential counter) provides efficient traceability without central coordination. The modular service architecture separating cryptographic primitives, cloud operations, and UI components establishes a maintainable codebase supporting incremental enhancement.

However, the current implementation exhibits critical limitations requiring remediation before production deployment. The LSB spatial-domain steganography completely fails under JPEG compression, limiting practical utility as social media platforms destroy watermarks through automatic re-compression. The static initialization vector creates deterministic encryption patterns enabling ciphertext analysis attacks. The absence of digital signatures prevents third-party verification workflows essential for legal and journalistic use cases. The direct SHA-256 password hashing lacks salt and iteration counts, making the system vulnerable to rainbow table and GPU-accelerated brute-force attacks. The CRC32 checksum detects accidental corruption but not intentional tampering by sophisticated attackers.

These limitations validate fundamental security engineering principles: spatial-domain steganography trades implementation simplicity for fragility under real-world transformations, necessitating frequency-domain techniques (DWT/DCT) for robust deployment; implementation shortcuts like zero IVs and direct hashing introduce exploitable weaknesses despite theoretically strong base algorithms; comprehensive content security requires both confidentiality mechanisms (encryption) and authenticity mechanisms (digital signatures) operating in concert; symmetric password-based authentication fundamentally differs from asymmetric public-key signatures in verification capabilities and appropriate use cases.

The feasibility analysis confirms that all proposed architectural enhancements are technically achievable within realistic development timelines. Digital signature integration leverages existing cryptographic libraries (pointycastle) with confirmed platform-secure key storage support on Android Keystore and iOS Secure Enclave. Frequency-domain watermarking prototypes demonstrate robustness improvements (94% JPEG compression survival vs. 0% for LSB) justifying 2-3x processing overhead. AES-GCM migration provides authenticated encryption eliminating CRC32 weaknesses while solving static IV issues through mandatory unique nonces. Cross-platform expansion to iOS, Windows, macOS, Linux, and Web leverages Flutter's shared codebase architecture with platform-specific testing and configuration.

The future development roadmap prioritizes security hardening (random IVs, PBKDF2 key derivation, AES-GCM) within 3-6 months, followed by digital signature integration and frequency-domain watermarking within 6-12 months, culminating in AI Detection API deployment and C2PA compliance integration as long-term objectives beyond 12 months. This phased approach addresses immediate cryptographic vulnerabilities while progressively building toward the complete proposed architecture incorporating third-party verification capabilities, robust watermarking surviving common transformations, and ecosystem integration supporting AI training dataset exclusion and industry-standard content authentication workflows.

From a broader impact perspective, TrueMark addresses critical contemporary challenges in digital content ownership, creator rights protection, and ethical AI training practices. However, technical protection mechanisms require complementary legal frameworks for effectiveness—the proposed AI Detection API enables platforms to identify protected images but cannot enforce rejection without regulatory mandates establishing binding compliance obligations. While TrueMark aims to democratize content protection previously available only to enterprises, deployment requirements including smartphone ownership, technical literacy, and workflow adaptation may inadvertently exclude creators in developing regions and elderly users, potentially exacerbating rather than reducing digital protection disparities.

The project delivers substantial educational value by illustrating multidisciplinary security engineering integrating cryptographic protocols, image processing algorithms, mobile application development, cloud infrastructure management, and user experience design. The comprehensive documentation of implementation decisions, identified limitations, proposed remediation strategies, and lessons learned provides reference architecture for future researchers and practitioners developing mobile content protection systems addressing similar challenges in audio watermarking, video protection, document authentication, and other digital asset security domains.

**In conclusion,** TrueMark successfully achieves its primary objective of demonstrating client-side cryptographic image protection feasibility on consumer mobile platforms while identifying clear implementation pathways toward robust production deployment. The working Android application proves that complex security operations can be abstracted behind intuitive interfaces accessible to non-technical users, validating that managed runtime environments provide adequate performance for real-time cryptographic operations. Though current limitations constrain immediate public release, the proven feasibility of proposed enhancements, modular architecture supporting incremental development, and comprehensive documentation establish a solid foundation for continued evolution toward the complete proposed architecture. As generative AI capabilities expand and content authenticity concerns intensify across journalism, legal proceedings, academic research, and creative industries, the demand for practical creator protection tools will grow correspondingly. TrueMark's demonstrated viability, validated enhancement pathways, and documented lessons learned position it as both functional proof-of-concept and reference architecture for the next generation of mobile content authentication systems addressing real-world problems at the intersection of technology, creativity, intellectual property rights, and ethics in the digital age.

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